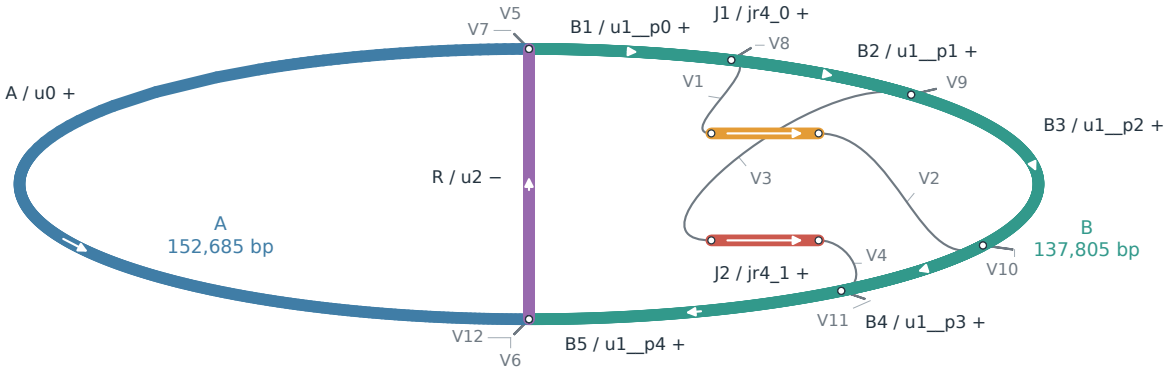
Same backbone, two supported alternatives, one nuclear-context branch

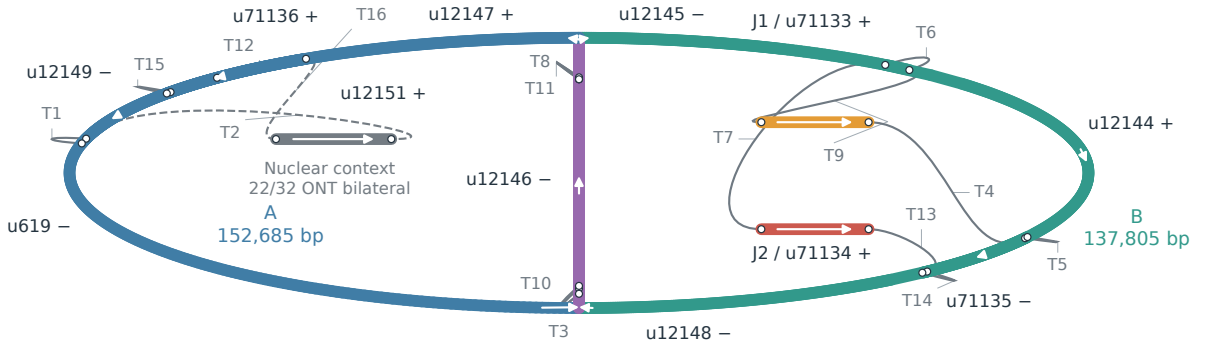
A backbone · B backbone · Shared R · J1 · J2 · Nuclear context

a OVASM repaired: 9 nodes / 12 links



Arrows follow the same homology orientation; +/− labels retain the original GFA strand.

b OATK: 12 nodes / 16 links



Arrows follow the same homology orientation; +/− labels retain the original GFA strand.

c Every GFA link has HiFi and ONT support; source still matters

OVASM link (original GFA orientation)	HiFi	ONT
V1 J1− → B1−	100	39
V2 J1+ → B3−	104	38
V3 J2− → B3+	337	27
V4 J2+ → B5+	333	27
V5 A− → R+	1608	697
V6 A+ → R−	1406	620
V7 B1− → R+	1332	595
V8 B1+ → B2+	1111	395
V9 B2+ → B3+	1123	379
V10 B3+ → B4+	1463	110
V11 B4+ → B5+	1480	110
V12 B5+ → R−	1487	675

OATK link (original GFA orientation)	HiFi	ONT
T1 u619+ → u12149+	1382	193
T2 u619+ → u12151−	46	37
T3 u619− → u12146−	1386	619
T4 u12144+ → u71133−	105	37
T5 u12144+ → u71135−	1474	110
T6 u12144− → u12145+	1171	392
T7 u12144− → u71134+	342	26
T8 u12145+ → u12146+	1259	594
T9 u12145− → u71133+	104	39
T10 u12146+ → u12148+	1481	675
T11 u12146− → u12147+	1526	698
T12 u12147+ → u71136+	1437	293
T13 u12148+ → u71134−	349	27
T14 u12148+ → u71135+	1511	110
T15 u12149+ → u71136−	1507	296
T16 u12151− → u71136+	49	37

J1 = u71133: 1,421 bp exact within jr4_0[319:1740).
J2 = u71134: 1,406 bp exact within jr4_1[81:1487).
These are shared local alternative connections, not missing MT sequence.

What differs after the repair?

Shared: exact A/B/R backbone and both HiFi + ONT supported local alternatives (J1/J2).

Different: node boundaries / overlap lengths; OATK alone retains u12151 (grey dashed branch).

u12151: 22/32 branch-compatible ONT records have ≥500 bp target-absent nuclear context on BOTH sides.

Strict context audit: ≥95% whole-query coverage and identity; HiFi 44/44 nuclear-aligned, 0/44 bilateral flanks.

Edge counts: 29,287 recruited HiFi + 2,126 ascertained complete ONT records; records may support several links.
Native protocol: k21, flank 2 kb, anchor 500 bp, diagonal tolerance 30 bp, spacing tolerance 5%; no depth/EC counts.
Colour follows exact homology, including mixed-colour OATK nodes. R copy mapping is ambiguous; it is shown once.
Schematic layout; crossings are not connections. Local support does not certify physical circles or transfer direction.